

WEEDON FIELD, AL, USA

WMO: 720363

Lat: 31.951N

Lon: 85.129W

Elev: 285

StdP: 14.55

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63872

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.5	(c) 28.1	(d) 9.5	(e) 9.0	(f) 36.1	(g) 13.9	(h) 11.3	(i) 37.3	(j) 16.5	(k) 60.0	(l) 14.4	(m) 59.0	(n) 2.0	(o) 340	(p) 0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	97.9	76.0	96.1	76.2	94.2	76.1	80.2	91.5	79.3	90.6	78.4	89.2	5.1	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	143.9	84.1	76.4	139.2	83.5	75.5	135.0	82.9	43.8	91.7	42.9	90.5	42.1	89.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.0	11.5	10.3	19.3	100.5	3.6	2.8	16.7	102.6	14.5	104.2	12.5	105.8	9.9	107.9
			17.5	82.1	3.7	1.1	14.8	82.9	12.6	83.5	10.6	84.2	7.9	85.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.9	48.7	53.2	59.5	66.1	73.8	81.0	82.4	82.3	77.8	67.3	56.6	52.6		
	DBStd	13.98	10.09	9.58	8.64	6.55	5.53	3.33	2.87	3.19	4.89	7.84	8.25	9.80		
	HDD50	361	149	72	20	2	0	0	0	0	0	1	29	88		
	HDD65	1884	512	343	213	63	10	0	0	0	1	68	274	400		
	CDD50	6513	108	162	315	485	739	929	1005	1002	833	538	228	169		
	CDD65	2561	6	13	43	96	284	479	540	537	384	140	23	16		
	CDH74	25949	29	105	469	1144	2928	5014	5618	5529	3507	1341	191	74		
	CDH80	12245	0	15	106	351	1305	2657	2876	2789	1653	456	28	9		
(14) Wind	WSAvg	3.5	4.1	4.2	4.6	4.3	3.7	3.0	2.8	2.8	2.9	3.2	3.2	3.5		
(15) Precipitation	PrecAvg	55.0	5.1	4.6	6.3	4.0	3.3	5.4	6.3	4.4	3.9	2.8	4.6	4.0		
	PrecMax	67.2	12.0	9.3	12.7	7.8	5.7	10.3	20.1	9.3	13.3	7.0	11.6	7.4		
	PrecMin	41.1	1.1	1.1	0.2	1.5	0.2	2.0	2.5	1.7	1.2	0.0	1.3	1.9		
	PrecStd	7.5	2.5	2.1	3.1	1.8	1.4	2.5	3.9	1.8	3.1	2.1	2.8	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.5	82.6	86.8	90.0	95.9	100.5	99.3	100.7	97.4	91.3	83.6	80.7		
		MCWB	66.0	68.3	65.9	68.9	71.0	74.8	77.4	77.5	74.3	72.7	68.5	67.7		
	2%	DB	73.2	77.2	83.1	86.9	92.2	97.5	96.9	97.5	94.5	88.3	79.0	75.5		
		MCWB	64.0	64.2	65.7	67.6	71.7	75.4	77.4	76.7	74.1	70.8	65.7	65.2		
	5%	DB	69.1	73.2	79.6	84.2	89.8	95.3	94.9	95.2	91.9	85.3	75.7	71.7		
		MCWB	62.6	61.8	64.3	66.5	71.0	75.8	77.3	77.1	74.4	69.4	63.6	65.1		
	10%	DB	65.3	69.4	75.5	81.2	87.4	92.9	92.8	92.8	89.5	82.0	72.1	68.2		
		MCWB	59.5	60.8	61.2	64.8	70.4	75.3	77.1	76.9	74.4	67.7	62.6	63.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.5	70.5	69.9	73.7	76.7	80.3	81.2	81.5	79.3	77.7	73.6	71.3		
		MCDB	72.6	78.6	77.9	83.0	85.0	92.0	92.3	93.2	88.8	83.7	76.6	74.9		
	2%	WB	66.3	67.5	68.2	71.4	75.0	79.0	80.1	80.2	78.1	75.4	70.3	68.7		
		MCDB	70.7	73.7	77.1	80.4	84.3	91.4	91.6	91.5	88.1	82.1	74.6	72.2		
	5%	WB	64.0	65.4	66.6	69.7	73.7	78.0	79.2	79.4	77.2	73.5	67.5	66.6		
		MCDB	68.4	71.0	75.8	78.1	83.8	90.2	90.8	90.2	87.0	80.1	72.8	70.3		
	10%	WB	61.2	62.6	64.7	68.0	72.6	77.1	78.3	78.4	76.1	71.4	64.5	64.2		
		MCDB	65.1	67.9	73.1	76.5	82.9	88.6	89.2	88.6	84.5	78.0	70.2	68.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.9	24.3	25.8	26.1	25.2	22.8	21.2	20.8	22.0	25.3	26.0	21.8		
		MCDBR	24.0	27.1	28.6	29.4	28.2	26.7	23.9	24.6	25.1	27.1	28.2	24.0		
	5% WB	MCWBR	15.6	15.4	13.4	12.5	10.8	8.7	8.1	8.1	9.2	12.3	15.8	14.9		
		MCWBR	20.3	21.4	23.7	22.0	22.3	22.7	21.2	20.8	20.6	20.0	20.5	18.5		
(40) Clear-Sky Solar Irradiance	taub		0.326	0.329	0.355	0.387	0.435	0.480	0.516	0.512	0.427	0.373	0.350	0.329		
	taud		2.522	2.509	2.448	2.354	2.257	2.190	2.097	2.103	2.329	2.454	2.464	2.527		
	Ebn at Noon		285	295	294	287	272	259	249	248	267	276	273	278		
	Edh at Noon		27	30	35	39	44	47	51	50	38	32	29	26		
(44) All-Sky Solar Radiation	RadAvg		923	1105	1472	1824	2009	1941	1913	1754	1560	1339	1036	796		
	RadStd		54	132	123	140	148	148	118	109	131	156	89	84		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+301	+196

Nomenclature: See separate page